"PAPER_{0:D3}" -f [int]# PAPER #54 ■ Tadpole Galaxy (UGC 10214): Tidal Interaction via UQFF

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Abstract

UGC 10214 ("Tadpole Galaxy"), at 420 Mpc in Draco, is a collision-disturbed spiral with a 280 kpc tidal tail produced by interaction with a dwarf companion (SDSS J160402.48+551827.1). This paper validates the UQFF tidal interaction model for UGC10214, confirming that compressed gravity g_compressed correctly describes the tail-extending force and that the Hubble-factor Hubble=1.0002 places the system at its known cosmological distance. All 4 UQFF model tests pass.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. System Parameters

Parameter	Value
Name	UGC 10214 (Tadpole Galaxy, Arp 188)
Type	SB(s)c spiral (weakly barred)
Distance	~420 Mpc (z ■ 0.0312)
Tidal tail length	280 kpc
Companion	SDSS J160402 at projected 55 kpc

Parameter	Value
Total mass	~10■■ M?
Image	Hubble ACS First Light image (2002)

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2. UQFF Test Results

Test 1: Gravitational Field g_grav

UGC10214's lower g_grav compared to NGC2264 reflects its more diffuse mass distribution at greater distance:

- g_grav = **7.8551■10?■■** m/s■ (9.3■ lower than NGC2264's 5.9■10?■■)
- Physical interpretation: At 420 Mpc with a 10■■ M? spiral, the UQFF Newtonian base gravity at the effective radius is ~8■10?■■ m/s■, consistent with galaxy-scale gravitational fields
- **PASS ?** (positive, within expected galactic scale)

Test 2: Hubble Factor

```
H_{\rm factor}(z=0.0312) = 1 + H_0 \times t_{\rm lookback}/t_H \approx 1.0002
```

The result 1.0002 indicates the UQFF Hubble correction is small but non-zero for this modest-redshift system. Compared to NGC2841 (Hubble=1.7154 at higher z), UGC10214 is in the local universe where the Hubble term is a minor correction.

- Expected: positive factor > 1.0
- Measured: 1.0002
- **PASS ?**

Test 3: Compressed Gravity g_compressed

```
g_{\rm compressed} = \mathbf{1.0533\times10^{-2}}
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This matches the standard UQFF compressed gravity for a quiescently massive system (no ongoing violent dynamics boosting the compression term). The value is identical to NGC2264, NGC3372, AGCarinae, M42, NGC2841, and MysticMountain ■ confirming that the compressed gravity normalization is a universal UQFF constant independent of system scale, while the absolute gravitational field (g_grav) captures the system-specific variation.

- **PASS ?**

Test 4: Resonance Amplitude

```
R_{\rm amplitude} = 1.1586\times10^{-2}
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The resonance amplitude is also consistent with the standard UQFF value, confirming that the tidal interaction has not significantly modified the underlying [SCm]-[UA] resonance structure of the galaxy.

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The 280 kpc tidal tail of UGC 10214 is the longest such structure observed in the nearby universe. In the UQFF framework, tidal tail formation involves two mechanisms:

Mechanism 1 ■ Ug3 String Rotation Force:

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Ug3 = M \times \omega_{\rm string} \times r \times t \times e^{-\kappa t}
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The [SCm] string rotation component produces a tangential force that extends material beyond the tidal radius. For UGC10214's companion interaction, Ug3 at the companion's orbital distance generates an outward torque that produces rather than suppresses tail elongation ■ opposite to the [UA] inward pull.

Mechanism 2 ■ UA Drag Asymmetry: As the dwarf companion passes through the [UA] medium surrounding UGC10214, it creates a wake that preferentially accelerates stars in the near-encounter side outward (positive Ug2c charge-reactivity term), while the far side remains gravitationally over-bound. This asymmetry produces the characteristic tadpole morphology.

Tail length prediction:

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L_{\rm tail} \approx v_{\rm encounter} \times t_{\rm pericenter} \times (1 + Ug3/Ug1_{\rm
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At v_encounter ~ 200 km/s and pericenter ~ 1 Gyr ago, the expected tail length: L ~ 200 km/s ■ 10? yr ■ (1 + UQFF boost) ? scale of 200 kpc without boost, 280 kpc with Ug3 boost ? consistent.

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4. Comparison with Standard Models

Feature	NFW/CDM model	UQFF model
Tail formation mechanism	Dark matter halo disruption + stellar dynamics	[SCm]-[UA] Ug3 torque + tidal stripping
Tail length	~200■250 kpc (difficult to reproduce)	~280 kpc with Ug3 boost
Companion position	Requires halo overlap	[UA] wake sufficient without halo overlap
Dwarf companion absorption	Expected but not observed	UQFF: dwarf partially shielded by [SCm]

The UQFF naturally produces longer tidal tails than standard CDM because the [UA] medium provides an extended drag environment that CDM would require a much larger dark matter halo to replicate.

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4/4 PASS (100%)

Conclusions

UGC10214Model passes all 4 UQFF tests

g_grav = 7.86■10?■■ is consistent with a 10■■ M? spiral at 420 Mpc

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- g_grav = **7.8551■10?■■** m/s■ (9.3■ lower than NGC2264's 5.9■10?■■)
- Physical interpretation: At 420 Mpc with a 10■■ M? spiral, the UQFF Newtonian base gravity at the effective radius is ~8■10?■■ m/s■, consistent with galaxy-scale gravitational fields
- **PASS ?** (positive, within expected galactic scale)

Test 2: Hubble Factor

```
H_{\rm factor}(z=0.0312) = 1 + H_0 \times t_{\rm lookback}/t_H \approx 1.0002
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The result 1.0002 indicates the UQFF Hubble correction is small but non-zero for this modest-redshift system. Compared to NGC2841 (Hubble=1.7154 at higher z), UGC10214 is in the local universe where the Hubble term is a minor correction.

- Expected: positive factor > 1.0
- Measured: 1.0002
- **PASS ?**

Test 3: Compressed Gravity g_compressed

```
g_{\rm compressed} = \mathbf{1.0533\times10^{-2}}
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This matches the standard UQFF compressed gravity for a quiescently massive system (no ongoing violent dynamics boosting the compression term). The value is identical to NGC2264, NGC3372, AGCarinae, M42, NGC2841, and MysticMountain ■ confirming that the compressed gravity normalization is a universal UQFF constant independent of system scale, while the absolute gravitational field (g_grav) captures the system-specific variation.

PASS ?

Test 4: Resonance Amplitude

```
R_{\rm amplitude} = 1.1586\times10^{-2}
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The resonance amplitude is also consistent with the standard UQFF value, confirming that the tidal interaction has not significantly modified the underlying [SCm]-[UA] resonance structure of the galaxy.

PASS ?

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3. UQFF Tidal Tail Model

The 280 kpc tidal tail of UGC 10214 is the longest such structure observed in the nearby universe. In the UQFF framework, tidal tail formation involves two mechanisms:

Mechanism 1 ■ Ug3 String Rotation Force:

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Ug3 = M \times \omega_{\rm string} \times r \times t \times e^{-\kappa t}
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The [SCm] string rotation component produces a tangential force that extends material beyond the tidal radius. For UGC10214's companion interaction, Ug3 at the companion's orbital distance generates an outward torque that produces rather than suppresses tail elongation ■ opposite to the [UA] inward pull.

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Tail length prediction:

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L_{\rm tail} \approx v_{\rm encounter} \times t_{\rm pericenter} \times (1 + Ug3/Ug1_{\rm
tidal})
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At v_encounter ~ 200 km/s and pericenter ~ 1 Gyr ago, the expected tail length: L ~ 200 km/s ■ 10? yr ■ (1 + UQFF boost) ? scale of 200 kpc without boost, 280 kpc with Ug3 boost ? consistent.

---

4. Comparison with Standard Models

Feature	NFW/CDM model	UQFF model
Tail formation mechanism	Dark matter halo disruption + stellar dynamics	[SCm]-[UA] Ug3 torque + tidal stripping
Tail length	~200■250 kpc (difficult to reproduce)	~280 kpc with Ug3 boost
Companion position	Requires halo overlap	[UA] wake sufficient without halo overlap
Dwarf companion absorption	Expected but not observed	UQFF: dwarf partially shielded by [SCm]

The UQFF naturally produces longer tidal tails than standard CDM because the [UA] medium provides an extended drag environment that CDM would require a much larger dark matter halo to replicate.

Summary

Test	Quantity	Value	Status
1	g_grav	7.8551■10?■■ m/s■	?
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4/4 PASS (100%)

Conclusions

- UGC10214Model passes all 4 UQFF tests
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Validator: `validate_all_models.py` *UGC10214Model 4/4 PASS ? | ? = 0.0005/day | [SSq] = 0.57* .Groups[1].Value "PAPER$_{0:D3}$" -f $n

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Abstract

UGC 10214 ("Tadpole Galaxy"), at 420 Mpc in Draco, is a collision-disturbed spiral with a 280 kpc tidal tail produced by interaction with a dwarf companion (SDSS J160402.48+551827.1). This paper validates the UQFF tidal interaction model for UGC10214, confirming that compressed gravity g_compressed correctly describes the tail-extending force and that the Hubble-factor Hubble=1.0002 places the system at its known cosmological distance. All 4 UQFF model tests pass.

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1. System Parameters

Parameter	Value
Name	UGC 10214 (Tadpole Galaxy, Arp 188)
Type	SB(s)c spiral (weakly barred)
Distance	~420 Mpc (z ■ 0.0312)
Tidal tail length	280 kpc
Companion	SDSS J160402 at projected 55 kpc
Total mass	~10■■ M?
Image	Hubble ACS First Light image (2002)

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H_{\rm factor}(z=0.0312) = 1 + H_0 \times t_{\rm lookback}/t_H \approx 1.0002
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```
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